KLE Technological University | Creating Value, Leveraging Knowledge

Department of Computer Application

CodeFest 2026

Domains and Problem Statements

1. Full-Stack Web Applications x E-Commerce

Scenario: Small enterprises in a city offer their products via their shops and social media. These enterprises have difficulties in reaching out to more customers, handling their orders and competing against big e-commerce players. The need arises for creating an accessible online marketplace which allows small enterprises to create an online identity and handle their inventories and orders.

Problem Statement: Develop and design a digital marketplace that helps local sellers create an online presence for themselves, manage their products, and reach their buyers through one single platform. The application should allow sellers to register, add and manage their products, track inventory, process orders, and monitor sales; while customers should be able to find local businesses, shop around for the products available, compare their options, and make purchases with ease. The teams are advised to think creatively about how to help buyers find local products, provide personalized recommendations, analyze seller performance, and engage with the customer.

2. AI-Powered Web Applications x Job Readiness

Scenario: Many students, recent graduates, and job applicants have problems carrying themselves well in interviews because of lack of adequate practice, proper feedback, and awareness of their weaknesses and strengths. Traditional interview practice sessions demand the availability of expert interviewees, which makes practice hard to come by. On the other hand, interview practice websites available on the market could lack dynamic interaction since questions and answers provided could be general and not suited to the individual's level of competency, expertise, targeted position, or performance in an interview. There is a need for an intelligent interview practice website that would allow users to gain more experience and practice interviewing through simulations.

Problem statement: Build a web-based application that makes use of AI to prepare individuals for interviews by creating an interactive, personalized mock interview experience. Such an application should allow users to choose a particular target job role, experience level, interview type, and level of difficulty. Relevant interview questions should be asked; interactive interviews should be conducted; user answers should be analyzed and evaluated; and personalized feedback should be provided regarding their communication skills, technical knowledge, problem-solving capabilities, and overall quality of answers. Teams can feel free to experiment with novel ways of doing adaptive questions, resume-based interview personalization, voice-based interviewing, automated answer evaluation, performance analysis, and personalized learning.

3. Artificial Intelligence and Machine Learning x Fintech

Scenario: Personal finance management poses many challenges for students as a result of limited income, unstable allowances, financial requirements of education, subscriptions, and impulsive behavior. Students face many problems with managing their money as it is quite hard for them to follow up on all expenditures, form stable saving habits and learn how to increase their savings. There are many apps which can help people in monitoring their expenditures but there is a need for something more than just following all expenses. The application should be able to assist students in forming financial discipline and learning responsible saving.

Problem Statement: Create and design an AI enabled student personal finance management application that will assist students in managing their finances and optimizing their expenses. With this application, students will be able to manage their income and expenses, categorize expenses, create budgets, set saving goals and track their financial performance. This application will utilize the power of artificial intelligence to study the patterns in spending of users, predict future expenses, discover opportunities for savings and offer suggestions for management of allowances, earnings from doing odd jobs or any other source of income. We encourage teams to experiment with new techniques for intelligent budgeting, customized savings plans, habit management of finances, expenditure forecasting, goal-oriented finance management, financial dashboards and more. Teams can also create educational modules that will teach students about concepts like compound interest, inflation, savings for unexpected situations and responsible investments to make their money work for them..

4. Data Analytics and Visualization x Sports

Scenario: Historical data is abundant in terms of the Olympics as regards nations, players, games, and events. Each new year of Olympics is rich in information on the number of medals won, participants of the games, advantages of hosting the games, sporting performance of teams, and variations in their competitive strengths. Nonetheless, most often, historical data is only helpful in describing something that has taken place in the past. There is a necessity to find more effective ways of analyzing historical tendencies, determining the factors of the performance, and predicting the outcomes of the next year of the Olympics. The number of medals won by any nation can vary owing to the development of players, the number of participants, new games added, variations in the categorization of games, or advantages of hosting the event.

Problem statement: Design and build a predictive analytics and visualization platform for the Olympics that performs analysis on the historical data about Olympics in order to discover performance trends, to find out the variables affecting medal wins, and to predict country-wise and sport-wise performance for the coming Olympics. The platform would allow the user to look at past editions of the Olympics, compare countries and sports, analyze the correlations among related variables, and make predictions about the future edition of the Olympics. Teams need to do more than just show historical medal tables. They need to analyze the data, explain their prediction methodologies, test their model against historical Olympics, and make predictions.